

RAHUL RAVI VK

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SUMMARY

Graduate Robotics Engineering student specializing in **control systems and industrial automation**, with hands-on experience in **PLC/DCS-based process control**, **SCADA environments**, commissioning, and troubleshooting in industrial operations. Strong foundation in **PID control** and data-driven optimization with cross-functional collaboration.

EDUCATION

Master of Engineering in Robotics (Full-time Student)

Sep 2025 – May 2027

University of Maryland, A. James Clark School of Engineering, College Park, MD

GPA: 3.9 / 4.0

Bachelor of Engineering in Instrumentation & Control

Aug 2014 – May 2018

L.D. College of Engineering, Gujarat, India

GPA: 8.25 / 10.0

EXPERIENCE

Engineering Assistant (Instrumentation & Control) — Operations & Systems Support

Oct 2018 – Jul 2025

KRIBHCO Fertilizers Ltd, Gujarat, India

- Supported plant-wide **instrumentation and control systems** using **Honeywell Experion PKS with C300 DCS**, assisting with loop checks, alarm monitoring, and daily control room operations.
- Assisted with **Safety Instrumented System (SIS)** monitoring and interlock verification using **Honeywell Safety Manager**, supporting safe startup, shutdown, and abnormal condition handling.
- Worked alongside engineering, operations, and maintenance teams to support **predictive maintenance and operational workflows**, contributing to a **50% reduction in equipment downtime**.
- Supported system updates, alarm rationalization, and control logic modifications under supervision, contributing to a **15% reduction in unplanned process interruptions**.
- Analyzed equipment usage and inventory data to support **data-driven decision making**, helping reduce excess inventory levels by **approximately 40%**.

PROJECTS

Mobile Robot Design, Simulation, and Control (ROS 2)

- Designed a mobile robot using SolidWorks, simulated in Gazebo. Implemented autonomous point-to-point navigation using a Python-based proportional controller and integrated LiDAR sensing to develop a high-fidelity digital twin.

Optimal Controller Design for Mobile Robot Trajectory Tracking

- Developed a MATLAB-based control framework utilizing Genetic Algorithms for trajectory optimization and a gradient-descent-tuned adaptive PID controller, validated through Lyapunov stability analysis.

Kinematic Replication Using UR5 Manipulator and Web Interface

- Implemented a decoupled ROS 2 control architecture for a UR5 manipulator, integrating a web-based user interface with a Python backend via WebSockets to execute real-time inverse kinematics in simulation.

Maze Exploration Using DFS with ROS 2 Actions

- Developed a C++17 ROS 2 Action Server implementing a Depth-First Search (DFS) algorithm to enable asynchronous navigation with dynamic replanning in response to environmental changes.

Air Dryer Process Automation — Siemens PLC

- Designed and implemented sequencing logic, interlocks, alarms, and safety conditions on a Siemens PLC, followed by system commissioning, functional testing, and operator training support.

SKILLS

Control & Automation: PID Tuning, LQR, LQG, Kalman Filtering, PLC, SCADA, DCS (Honeywell C300), SIS

PLC Programming: Ladder Logic (basic), Structured Text (exposure), Siemens PLCs

Robotics & Simulation: ROS 2 (Humble), Gazebo, MoveIt, UR5, URDF/Xacro, RaspberryPi, Arduino, openCV

Programming & Tools: C++ (C++17), Python (NumPy, SciPy), MATLAB, Git, Linux/Ubuntu, SolidWorks

CERTIFICATIONS

Siemens — Basics of PLC & SCADA

Honeywell — Experion PKS C300 Fundamentals